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RESEARCH ARTICLE

Enhanced MPP Tracking in Partial Shading Conditions for Solar PV Systems: A Metaheuristic Approach Utilizing Projectile Search Algorithm

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ABSTRACT For solar Photovoltaic (PV) systems to function effectively amid dynamic conditions, it is imperative to achieve efficient power harvesting. Given the escalating demand for energy, the application of solar PV technology for electricity generation must be optimized to ensure optimal performance and cost-effectiveness. Maintaining an ample supply of power at the lowest cost is crucial in this regard. Partial Shading Conditions (PSCs) significantly reduce power transfer efficiency in solar PV systems, potentially leading to the formation of hotspots within the solar array. While the insertion of bypass diodes can address this issue, it often results in multiple power peaks on the Power vs Voltage (P-V) curve characteristics, complicating the process of maximum power tracking. To overcome this challenge and alleviate the computational load on the microcontroller, the application of metaheuristic techniques for Maximum Power Point Tracking (MPPT) proves beneficial. However, due to the distinctive operational characteristics of metaheuristic algorithms, continuous research in this domain is essential. Addressing the need for effective Maximum Power Point (MPP) capture in diverse partial shading scenarios, this study introduces a novel MPPT technique based on the Projectile Search Algorithm (PSA). The PSA, inspired by the projectile motion of physical objects, is a metaheuristic optimization algorithm tailored for solving optimization problems by simulating projectile motion within a search space to identify optimal solutions. To assess the performance of the proposed PSA-based approach, comparisons are made with Jaya, Cuckoo Search, Particle Swarm Optimization (PSO) and Perturb and Observe (P&O) algorithms. Real-time validation using the Typhoon Hardware in the Loop (HIL)-402 emulator is employed to verify the suggested approach, and MATLAB/Simulink software is utilized for evaluation. A comprehensive analysis of the results, considering tracking time, power tracking efficiency, and power fluctuations, demonstrates the superior performance of the proposed algorithm compared to existing methodologies.

INDEX TERMS Projectile search algorithm (PSA), maximum power point tracking, partial shading condition, metaheuristic algorithms, photovoltaic (PV), optimization.

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I. INTRODUCTION

Environmental problems, including polar ice cap melting, climate change, and global warming have become worse during the previous ten years, mostly as a result of human

activity [1]. Nowadays, fossil fuels are used more quickly as the primary cause of power production due to the rising demand for energy. Long-term fossil fuel usage cannot be supported. These present dangers have increased interest in using solar PV systems to provide renewable, clean energy. Furthermore, solar PV systems are inexpensive, long-lasting, low-maintenance, and suitable for any size electrical grid due to their lack of moving components [2].

The PV system exhibits a non-linear P-V and I-V characteristic curve with a single MPP when the irradiation is uniform. The PV system will only function at one point, regardless of the MPP, when a fixed load is connected in parallel. Only variations in solar irradiance or the surrounding temperature impact lower power generation. A DC-DC converter with an MPPT controller is often connected to the PV system to allow maximum power generation via modifying the equivalent circuit resistance based on Thevenin's theory. Nevertheless, the MPPT can become more complex when partial shading effects on PV panels are brought on by moving clouds, buildings, trees, bird droppings, and dust. These effects can also result in multimodal P-V curves with several Local Maximum Power Points (LMPP) and one Global Maximum Power Point (GMPP) [3]. Over the last 10 years, increasing the efficiency of solar photovoltaic systems has proven to be quite challenging. Improving the tracking performance of algorithms of MPPT requires significant study since it may maximize efficiency without modifying any physical PV system components, is more affordable, and is simpler to deploy. There are several subcategories of MPPT approaches, including, Artificial Intelligence (AI), hybrid, metaheuristic, advanced and traditional techniques. Due to their straightforward implementation and low cost, several conventional methods, such as Incremental Conductance (INC) and Perturb and Observe (P&O), are frequently used in industrial applications [4]. The beginning particle location and step size have a significant impact on the tracking speed. Another issue with P&O approaches is oscillation around MPP, which results in a trade-off between efficiency and tracking speed. A researcher in [5] presented Proportional-Integral-Derivative (PID) based linear controller techniques. The non-linear P-V characteristic curve, however, is not compatible with the linear PID characteristic. Sliding Mode Controller (SMC) [6], a non-linear controller with improved stability against PV system change, is growing in favor as a means of overcoming the limitations of PID controllers. SMC, on the other hand, experiences noise, and control gain selection has a significant impact on performance. The approaches listed above demonstrate the ability to monitor single-peak PV curves but carry a significant risk of trapping the user at a nearby peak.

Numerous AI soft computing methods, including Artificial Neural Network (ANN) [7], [8] Fuzzy Logic Controller (FLC) [9], and Adaptive Neuro Fuzzy Inference System (ANFIS) [10], were put out to solve the PSC issue and address

the drawbacks of traditional MPPT techniques. Although these techniques based on AI offer quick convergence rates towards the GMPP, their implementations are frequently too difficult. Training data often affects how well ANN algorithms predict outcomes, and improper training techniques may result in overfitting problems [11]. Despite the fact that the proposed ANN-based MPPT algorithms in [8] can predict GMPP using temperature and sun irradiance data collected from several photovoltaic panels, these methods require an expensive sensor setup. Regarding FLC-based MPPT algorithms, in-depth knowledge of PV systems, such as membership functions, linguistic terms and fuzzy rules, was needed to define the controller parameters. Metaheuristic approaches inspired by many natural occurrences, such as physics, mathematical ideas, animal foraging behavior, human activities, and natural evolution, are increasingly being used to solve MPPT issues [12]. PSO [13], [14], [15], [16], [17], salp-swarm optimization [18], [19], [20], Bat Algorithm (BA) [21], [22], Atomic Orbital Search (AOS) [23], Butterfly Optimization Algorithm (BOA) [24], Jaya [25], Flying Squirrel Search Optimization (FSSO) [26], Firefly Algorithm (FA) [27], Improved Grey Wolf Optimizer (IGWO) [28], [29], Low Burden Narrow Search (LBNS) [30], Cuckoo Search (CS) [31], and Most Valuable Player Algorithm (MVPA) [32], energy valley optimizer [33] are some common and recent metaheuristic techniques for finding MPPT in PSC. Compared to conventional methods, these stochastic metaheuristic-based MPPT algorithms are more capable of escaping from LMPPs and tracking GMPPs with almost no steady-state oscillations. Additionally, these MPPT algorithms based on metaheuristic are less impacted by solar panel degradation and don't rely on PV system specifications [34], [35]. Despite the advantages, it is difficult to create dependable, useful PSO-based MPPT algorithms for real-time applications because of some persistently difficult problems. For instance, factors like population size, population beginning locations, inertia weight, and acceleration coefficients have a significant impact on the tracking capabilities of various based on PSO MPPT algorithms. These parameter tuning procedures were most often carried out on a trial-and-error basis, which tends to increase complexity and computational cost. Poor tracking results may occur from improper initialization of these parameters, which might cause early or sluggish convergence towards LMPPs [36]. Another interesting strategy to achieve the correct equilibrium between extraction and exploration processes while addressing MPPT issues is the hybridization of two separate kinds of MPPT approaches [37], [38], [39], [40]. While PSO was hybridized with other approaches in [41], [42], and [43] to address the drawbacks of the individual MPPT approach and produce a cooperative result that enhances tracking efficiency. Finding the ideal algorithmic combinations is difficult, and an excessive number of sensors and parameters in an MPPT algorithm may result from a poorly planned hybridization process.

Furthermore, advanced MPPT control methods are a broad category of approaches with the goal of improving solar PV systems' efficacy and efficiency. Innovative strategies like Centralized MPPT technique [44] based on Sliding Mode Control and Distributed Sliding Control (DSC) [45], which uses distributed control algorithms to maximize PV array performance across a range of operating situations, have been the subject of recent investigations. The ability of Hybrid Time-Frequency Analysis (HTA) approaches [46] to examine the temporal and spectral properties of solar irradiance data has also attracted attention. This capability allows for more precise monitoring of the highest power point. Furthermore, studies in Neuro-Linguistic (NL) [47] and Reinforcement Learning (RL) [48] methodologies have shown potential for real-time MPPT strategy adaptation depending on system dynamics and contextual parameters. Additionally, Unsupervised Cluster Method (UCM) [49] approaches have surfaced, providing new means of determining ideal operating conditions using data-driven clustering methods. These cutting-edge methods represent a change towards more complex and adaptable MPPT control schemes that may optimize energy output and improve solar PV systems' overall performance.

A cost-effective algorithm with reliable performance is still needed, according to current MPPT research.

This study presents Projectile Search Algorithm (PSA), a novel metaheuristic algorithm configured specifically to solve the MPPT problem in solar PV systems. The unique characteristics of the algorithm are explained in detail, emphasizing its applicability to MPPT applications. The key contribution of the work includes

1. The simulation and the real-time results from the Typhoon Hardware-in-the-Loop (HIL) 402 have been presented with various near-practical case studies.
2. A thorough performance evaluation of the proposed algorithm based on various performance parameters has been done.
3. A comparison has been done with well-established and standard algorithms like PSO, Cuckoo Search, and Jaya, three popular nature-inspired algorithms.

Furthermore, the suggested MPPT method's salient features include—low computation cost, quick convergence, and little reliance on random number generation—are emphasized, bolstering its effectiveness and suitability for use in actual solar PV systems.

The sections are categorized as follows, according to the following groupings: Section II details the formulation of solar photovoltaics. The next part, Section III, provides information on MPPT and PSC, as well as their influence on solar photovoltaic arrays. Section IV provides a comprehensive discussion of the suggested PSA. Section V illustrates the performance evaluation of the MPPT for various PSCs using the PSA approach. Section V also presents the results of the PSA's real-time comparison with other cutting-edge algorithms, such as Jaya, PSO, and Cuckoo search algorithms. Finally, Section VI offers an overview of the study paper's

results as well as insights into the work's potential future directions.

II. PV FRAMEWORK FORMULATION

The fundamental component of a PV system is a PV cell. Figure 1 illustrates how solar PV cells are connected in series to make solar PV modules. A semiconductor device known as a photovoltaic cell uses the photovoltaic process to transform solar energy into electrical energy. When the daylight incident photon energy on the PV module exceeds the energy gap, the electrons from the PV cell begin to proceed and create an electric flow. Figure 2 displays a model of a similar solar cell Single Diode Model (SDM).

PV modeling is crucial for accurately simulating the performance and behavior of PV systems under various conditions, enabling the optimization of energy generation and system design. Over the years, numerous PV models have been introduced in the literature to accurately and simply replicate the PV characteristic curves as outlined in manufacturer datasheets. Depending on their derivation, these PV models can be grouped into three main categories: circuit-based, analytical-based, and empirical-based models [50], [51]. Equivalent-circuit-based models are among the most widely used methods for representing the behavioral characteristics of the I–V curve. These models employ conversion principles to describe a typical PV panel and can be categorized based on the number of diode components into single-diode, double-diode, or triple-diode models. Each model varies in complexity and accuracy: the single-diode model offers a balance between computational simplicity and sufficient accuracy for many applications [52], whereas the double-diode model provides more detailed representations at the cost of increased complexity [53]. The single-diode model is employed in this paper due to its effectiveness in capturing the essential characteristics of PV cells, such as the current-voltage relationship, with relatively low computational demands. This model is particularly advantageous for large-scale simulations and real-time applications where efficiency and speed are critical.

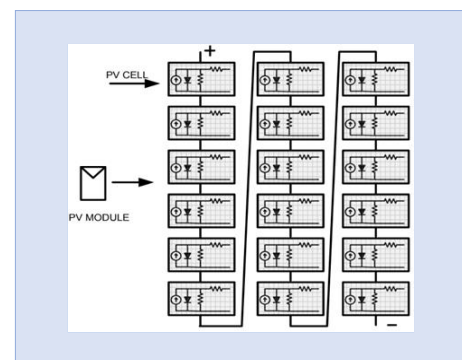


FIGURE 1. Sequential connections of solar PV cells to create a solar PV module.

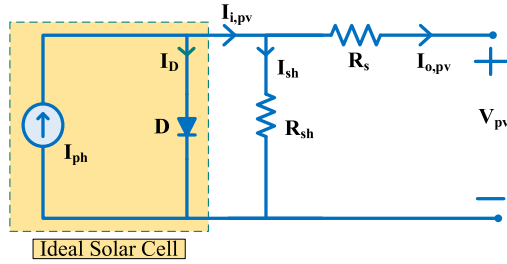


FIGURE 2. Solar cell Single Diode Model (SDM).

Further the parameter extraction and comprehensive modelling of solar PV systems by means of the Archimedes Optimization Algorithm (AOA) and Tabu Search Optimization has been mentioned in [54] and [55].

Where I_D stands for the reverse current in a PV cell in its simplified electrical circuit. I_{ph} stands for the photon-offered current, whose importance depends on the PV cell ability to gather radiation, the ambient temperature, and the light intensity rays. I_{sh} stands for the shunt current, which is the bypass resistor (R_{sh}) direct byway. Its creation is also caused by the metal extension overflowing onto the metal electrode and the edge of the battery. $I_{i,pv}$ represents the PV cell input terminal current. R_s stands for the resistor with a series connection that depends on the contact's and the material resistance. $I_{o,pv}$ is the shorthand for the output terminal current of the PV cell.

The output current ($I_{o,pv}$) of a PV cell is usually found using Kirchhoff's current law, which is computed as shown in equation (1):

$$I_{o,pv} = I_{ph} - I_D - I_{sh} \quad (1)$$

Equation (2) provides the mathematical expression for the operating temperature and solar irradiation level. These factors are what cause the PV cell's current to be generated when solar light triggers it [56].

$$I_{ph} = I_{SC} + \left[K_i \times (T_c - T_r) \times \left(\frac{G}{G_{STC}} \right) \right] \quad (2)$$

where I_{SC} shows the PV cell short-circuit current under Standard Test Conditions (STC); K_i represents, in relation to the temperature coefficient, the cell short-circuit current(A/K); G represents the amount of solar insolation, measured in watts per square meter, that is received by the PV cell's face area. T_r and T_c stand for the cell reference temperature and operating temperature, respectively.

Likewise, I_O denotes the reverse saturation current that differs per the condition of the environment which can be mathematically evaluated as shown in equation (3):

$$I_O = (I_{O,STC}) \times \left\{ \left(\frac{T}{T_{STC}} \right)^3 \times e^{\left(\left(\frac{qE_g}{kA} \right) \cdot \left(\frac{1}{T} - \frac{1}{T_{STC}} \right) \right)} \right\} \quad (3)$$

Equation (4) uses the Shockley equation to determine the current flowing through the diode.

$$I_D = I_O \times \left(\left(e^{\frac{q(V_{pv} + (I_{ph} \cdot R_s))}{V_T}} \right) - 1 \right) \quad (4)$$

Here, V_T stands for thermal voltage, while V_{pv} is the solar cell's output voltage on the load side. Equation (5) illustrates the current calculation via the shunt resistor using Ohm's law.

$$I_{sh} = \frac{V_{pv} + (I_{ph} \cdot R_s)}{R_{sh}} \quad (5)$$

Hence, equation (6) and (7) demonstrates the characteristic algebraic equation of a solar cell as follows [56]:

$$I_{o,pv} = I_{ph} - \left(I_O \times \left(\left(e^{\frac{q(V_{pv} + (I_{ph} \cdot R_s))}{V_T}} \right) - 1 \right) - \left(\frac{V_{pv} + (I_{ph} \cdot R_s)}{R_{sh}} \right) \right) \quad (6)$$

where,

$$V_T = \frac{(N_{se} \times A \times k \times T)}{q} \quad (7)$$

where, N_{se} is the number of solar cells connected in series.

A. CHARACTERISTIC CURVES OF A SOLAR CELL

When a solar cell is in a short-circuit condition, the current flow is known as the short-circuit current. Conversely, the term open-circuit (V_{oc}) describes the optimal voltage, which represents the open-circuit electrical network of the solar cell. The maximum amount of power accessible corresponds to a specific combination of voltage and current, but neither the I_{SC} nor the V_{oc} contribute to power creation. This is even though there is such a combination.

Figures 3 and 4 show the characteristics of a PV cell's Current versus Voltage (I-V) and P-V graphs for variations in insolation level at constant temperature, respectively. It is possible to draw the following conclusions from Figures 3 and 4: An increase in solar irradiation causes a modest rise in the open-circuit voltage of the PV cell. On the

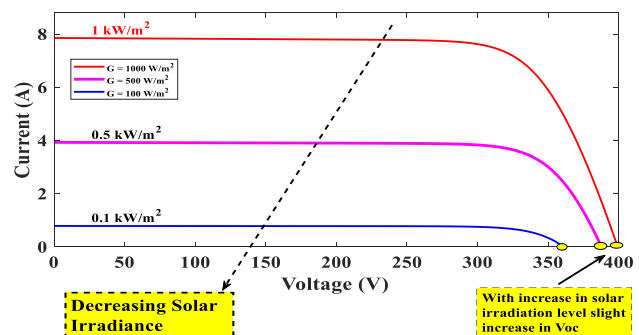


FIGURE 3. The I-V characteristic curve of a solar cell shows how solar irradiation varies at a given temperature.

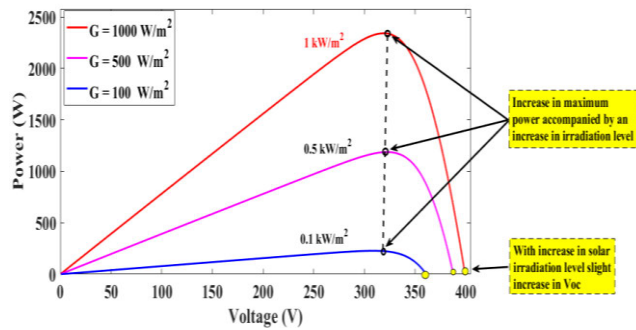


FIGURE 4. Characteristic curve of a solar cell P-V for changes in solar irradiance at fixed temperature.

other hand, the short-circuit current flowing through the PV cell has significantly increased. As a result, the irradiation level has a direct influence on the yield power of a PV cell at a constant temperature.

III. MAXIMUM POWER POINT TRACKING (MPPT)

Because of the nonlinear nature of the PV array, MPPT methods are required. The maximum practicable output from the PV module in various scenarios is determined using MPPT algorithms by correlating the voltage at the output of the PV arrays to the voltage of the battery [57]. AC is converted from the DC input data of the PV modules by the DC-to-DC converter. It then switches to various DC currents and voltages to accurately support the solar panels. The MPP is the voltage at which a PV system yields its maximum output. Figure 5 illustrates the basic block structure of the MPPT architecture.

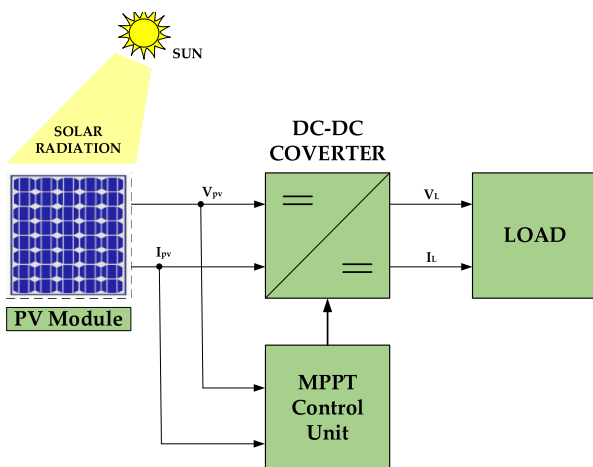


FIGURE 5. Diagrammatic representation of an MPPT framework block.

A. PARTIAL SHADING CONDITIONS (PSC)

Clouds, dust, tree fragments, building structures, and other factors all contribute to the shading problem. Figure 6 depicts a series-linked solar photovoltaic system with two PSC-covered modules. When installing a solar framework,

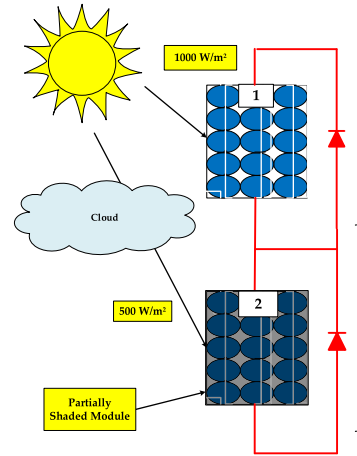


FIGURE 6. Solar PV array with two modules under PSC.

it is considered one of the most important considerations. In PSC, the PV array's shaded modules get less solar insolation than the exposed modules. As a result, an excessive quantity of the electric energy generated by the PV cells that are not shaded is absorbed by the hidden PV cells, which are then used to generate heat. In certain situations, the generated heat could potentially harm the shaded cells. Additionally, PSC caused several peaks to appear in the PV module's P-V characteristics. As a result, bypass diodes are used throughout the PV modules to lessen the load on the shaded cells, shielding the concealed cells from any potential damage [58].

Figures 7 and 8 depict the graphs of a PV module's current versus voltage and power versus voltage during PSC, respectively. Figure 7 displays a solar panel's two I-V characteristic curves with partial shade of 1000 m/s² and 500 m/s². The I-V characteristic illustrates how the solar panel's current flows as the voltage increases. The IV curve is displayed by the purple line at an irradiation value of 500 m/s², while the orange line shows the I-V curve for levels of irradiance of 1000 m/s². Up to a certain point, the voltage and current are proportionate. Past that threshold, the current saturates and becomes less sensitive to voltage increases. The point on the I-V curve where power is at its maximum is called the

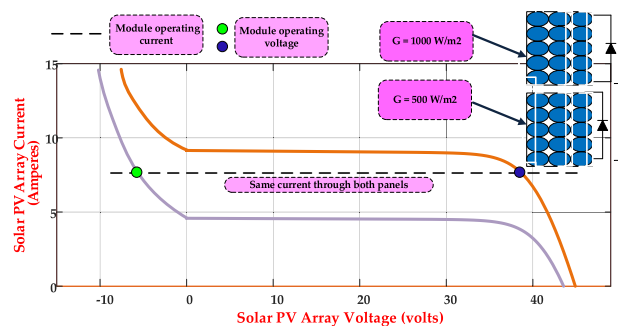


FIGURE 7. Solar cell I-V plot characteristics under the PSC.

MPP, and the point at which the solar panel is producing its maximum energy is called the saturation point. As the shade level rises, the panel's voltage output stays constant but its current output drops. Additionally, when the shading level rises, the I-V curve shifts to the right. This indicates that a greater voltage is needed at the panel to get the same amount of current.

Figure 8 illustrates the characteristic P-V curve of a partially shaded solar panel, highlighting how it may display distinct features under partial shade due to uneven lighting across the array. A solar PV array's P-V curve provides crucial insights into its behavior under various conditions, including partial shade. In typical, unshaded circumstances, the PV curve exhibits a distinctive form, with current increasing proportionally to voltage until reaching a peak known as the MPP.

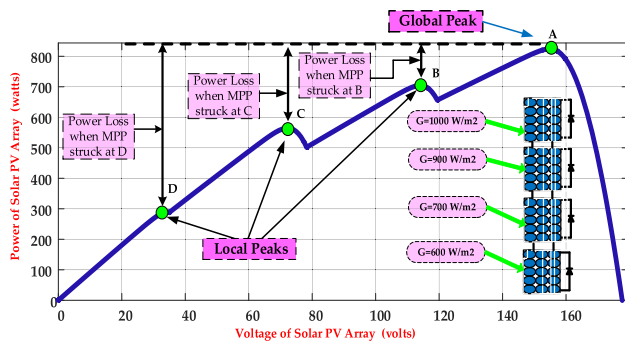


FIGURE 8. Solar cell P-V plot characteristics subjects to the PSC.

On a PV array, partial shading may result in numerous local MPPs appearing on the PV curve. Based on the specific I-V relations of the array, these are the points at which a shaded zone operates at maximum power output. The overall power generation might be limited to the lowest of these local MPPs for every shaded area of the array. There will be a local MPP assigned to each shaded section. This restriction occurs because the darkened regions function as resistors, resulting in voltage dips and reduced current flow. Because of this, the global MPP is the point on the IV curve at which the whole array operates at its highest possible power, while the local MPPs show the ideal operating locations for each shaded region. It considers the overall impact of all the shaded and unshaded areas. Under partial shade, it might be difficult to reach the global MPP because shading-related changes in voltage and current can cause the system to drift from its ideal state. Techniques include shade analysis, bypass diodes, and sophisticated MPPT algorithms that dynamically modify the array's current-voltage characteristics to steer the system towards the global MPP. Direct shadowing of a PV module leads to a decrease in power output. However, shading does not affect the fill factor or efficiency of PV modules [59]. In conditions when there is insulation, it is necessary to keep track of the maximum power that is accessible at a certain knee position. Because of the shade caused by passing

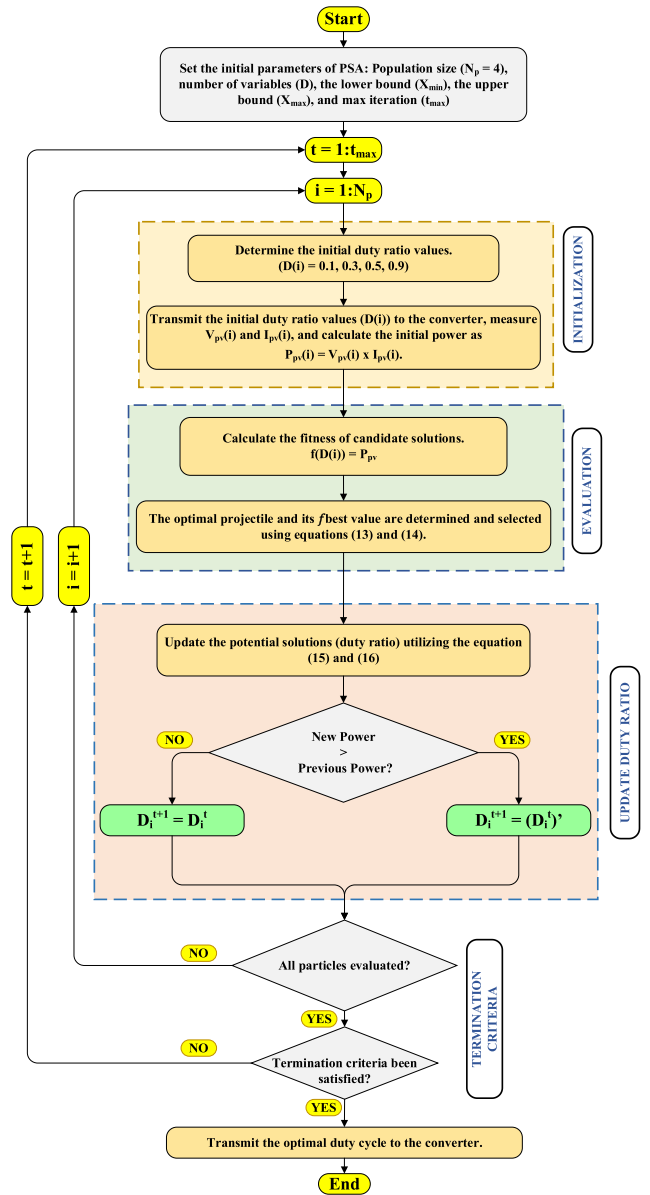


FIGURE 9. Flowchart of PSA for MPPT.

clouds, solar panels linked in series get varying amounts of light. Due to the hotspot development issue, the temperature of the shaded panel rises, potentially resulting in PV panel breakdown. The issue may be lessened using the bypass diodes [60]. Each panel is wired in parallel to bypass diodes. Hotspot generation is prevented during shading by avoiding the shaded panel.

IV. PROJECTILE SEARCH ALGORITHM (PSA) OPTIMIZATION

The PSA is a metaheuristic optimization technique that draws inspiration from an actual object's projectile motion. It is intended to identify the best answer to optimization issues by mimicking a projectile's motion in a search area. PSA

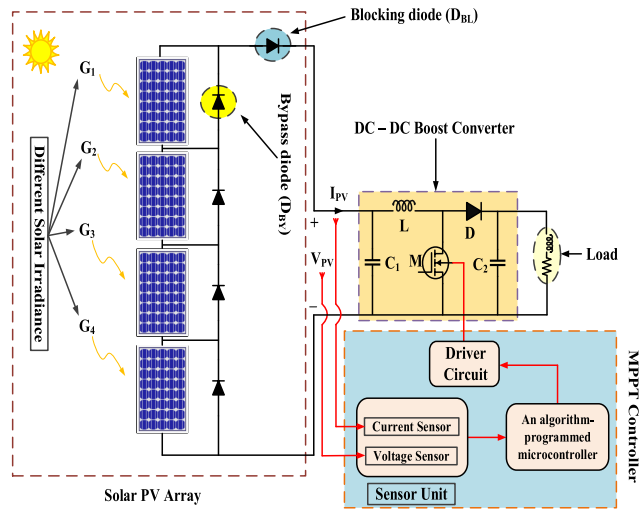


FIGURE 10. MPPT circuit realization in typhoon HIL.

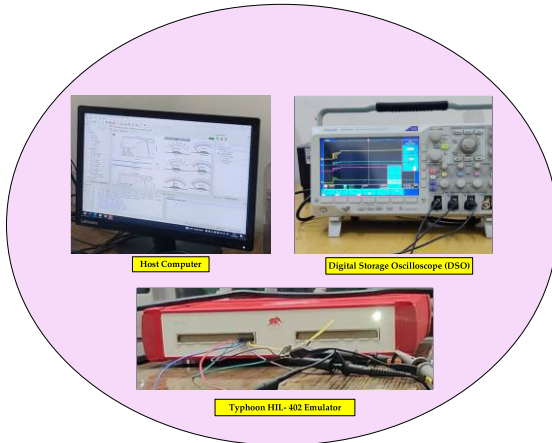


FIGURE 11. Computer serving as the host, Typhoon HIL-402 emulator for real-time output, and Digital Storage Oscilloscope (DSO) for graph visualization.

utilizes the concept of a projectile, which represents a potential solution to the optimization problem [61]. The projectile is launched at a set speed from a ground starting point, propelled in many directions while keeping its gravitational field constant, and then brought to rest on a target located at a surface. On the other hand, the PSA that has been presented does not emphasize the trajectory of a projectile that has been released at an angle. One of the most popular applications of this technology is the tracking of the projectile trajectory envelope. This envelope encompasses all the potential sites in the search space. The envelope of the projectile trajectory can approach places in the search space that are inaccessible to projectiles travelling from any beginning point and at any velocity [62]. When aiming a projectile towards a target from its origin point, having accurate knowledge about the trajectory of the projectile might be helpful. Each projectile is characterized by its position, velocity, and fitness value. The algorithm iteratively updates the positions

Algorithm 1 PSA Algorithm Pseudocode

Step	Description
1	Initialize: - Set the number of projectiles (N) - Initialize positions (P) for each projectile randomly within the search space - Initialize velocities (V) for each projectile randomly within the feasible range - Determine each projectile fitness value by applying the objective function to it
2	Set termination criteria: - Choose the maximum number of iterations (Max-Iterations) - Set the desired fitness value (TargetFitness)
3	Initialize the best position discovered so far (Best-Position) and set it to the position of the projectile with the highest fitness value
4	Repeat until termination criteria are met: Motion Update: - Based on each projectile's present location, velocity, and BestPosition, update its position and velocity - Use the motion update equations inspired by projectile motion Evaluate fitness: - Calculate the fitness value for each projectile based on the objective function Best Position Update: - Update BestPosition if a projectile discovers a better position with a higher fitness value Termination check: - Check if the termination criteria are met: - If the MaxIterations is reached, terminate the algorithm - If the BestPosition's fitness value is equal to or higher than the TargetFitness, terminate the algorithm
5	Return the BestPosition as the optimal solution

and velocities of the projectiles to search for the optimum solution.

A. PROJECTILE SEARCH ALGORITHM

The PSA technique starts with the population size (N_p), number of variables (D), lower limit (X_{min}) upper limit (X_{max}), and maximum iteration T_{max} [62]. The variables lower and upper boundaries are stated in equation (8) as:

$$X_{minj} = [X_{min1} X_{min2} \dots X_{minD}], X_{maxj} = [X_{max1} X_{max2} \dots X_{maxD}] \quad (8)$$

To create the initial population of projectiles as potential solutions, random values *randj* from the range [0, 1] are

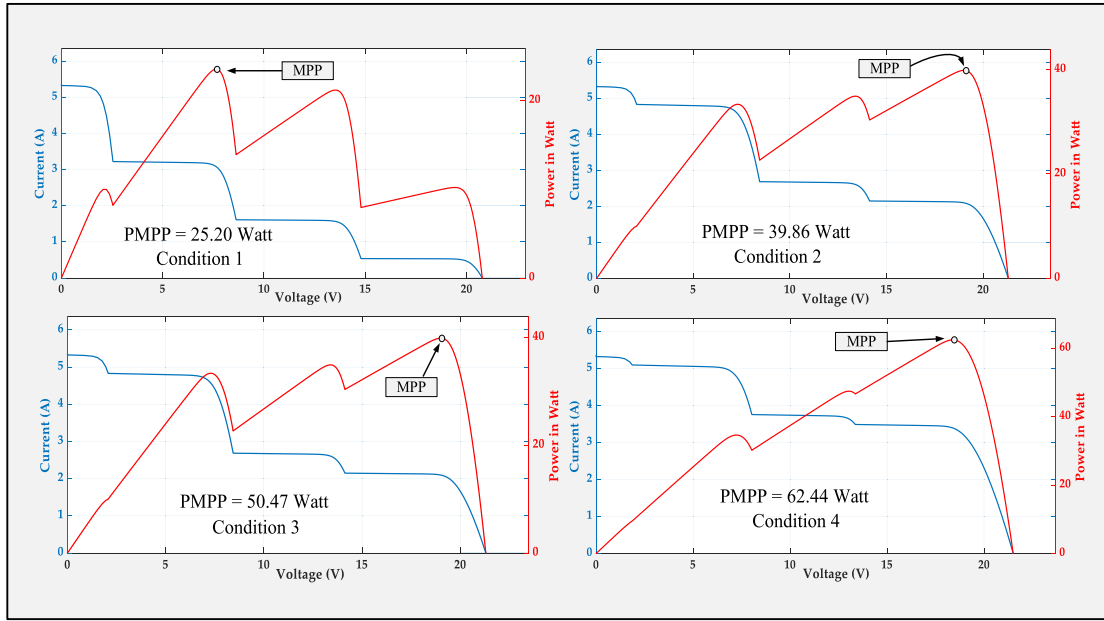


FIGURE 12. Four PV modules receive varying amounts of solar irradiation.

assigned to each border as follows in equation (9):

$$X_{i,j}^{I=1} = X_{minj} + rand_j, (X_{maxj} - X_{minj}),$$

$$i = 1, 2, \dots, N; j = 1, 2, \dots, D \quad (9)$$

The target population is formed at random by disturbing one randomly selected projectile with the differential between the other two randomly selected projectiles. The intended population is produced by using equation (10).

$$t_{k,j}^{(I)} = X_{a1,j}^{(I)} + 0.5 (X_{a2,j}^{(I)} - X_{a3,j}^{(I)}),$$

$$k = 1, 2, \dots, N; j = 1, 2, \dots, D \quad (10)$$

where the indices such as $a1$, $a2$, and $a3$ belongs to $\{1, 2, \dots, N\}$ randomly selected numbers that must be distinct from one another and from the base index.

Using (11) and (12), the following criteria are used to evaluate the targets and projectiles fitness values:

$$f_i^{(I)} = f(x_{i,j}^{(I)}) \quad (11)$$

$$f_k^{(I)} = f(t_{k,j}^{(I)}) \quad (12)$$

Then, by contrasting the fitness values of each $X_{i,j}^{(I)}$ and $t_{k,j}^{(I)}$, the best projectile $X_{best}^{(I)}$ and its best value $f_{best}^{(I)}$ are chosen using equation (13) and (14) logic:

$$X_{bestj}^{(I)} = \begin{cases} X_{i,j}^{(I)} & \text{if } \min(f(X_{i,j}^{(I)})) \leq \min(f(t_{k,j}^{(I)})) \\ t_{k,j}^{(I)} & \text{otherwise} \end{cases} \quad (13)$$

$$f_{best}^{(I)} = f(X_{bestj}^{(I)}) \quad (14)$$

where I is the succeeding generation produced by each iteration and consists of $1, 2, \dots, I_{max}$. Iterating the projectile

until it has the greatest fitness value is the primary PSA optimization method. The following equation (15) and (16) describes the projectile search paradigm for focusing on the target:

for $i = 1 : N$

for $j = 1 : D$

If $rand < 0.5$

$$X_{i,j}^{(I+1)} = X_{best1,j}^{(I)} - \alpha \left(\frac{\gamma(d_{i,j}^{(I)})}{\gamma'(d_{i,j}^{(I)})} \right)$$

Else (15)

$$X_{i,j}^{(I+1)} = X_{i,1}^{(I)} + \alpha \left(\frac{\gamma(d_{i,j}^{(I)})}{\gamma'(d_{i,j}^{(I)})} \right)$$

end

end

end (16)

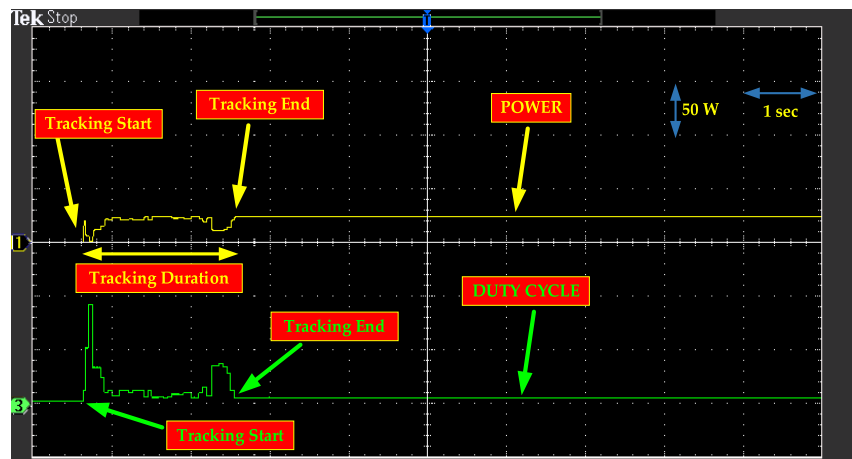
With a slope of m and a linear shape, the impact function in PSA optimization is represented by a hill in equation (17).

The function of impact is

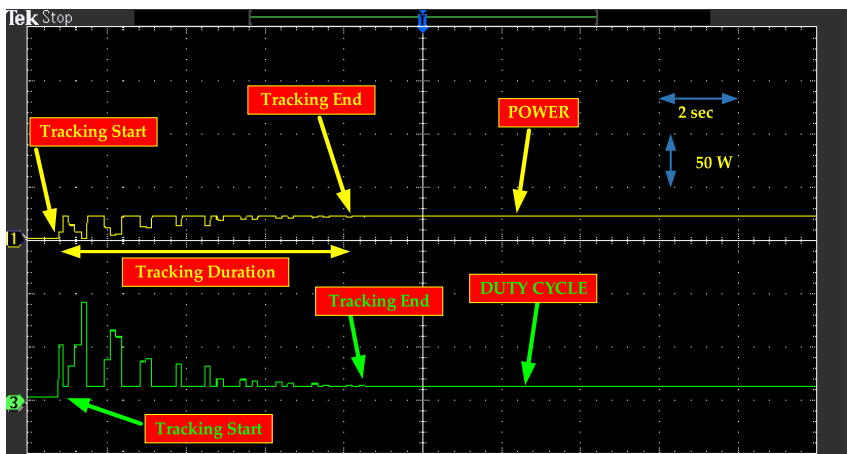
$$\psi(X) = m.X \quad (17)$$

Given the function of impact specification [63], the function (X) might be defined as shown in equation (18) and (19)

$$\gamma(X) = m.X - \frac{v^2}{2g} + \frac{g}{2v^2}x^2 \quad (18)$$



(a)

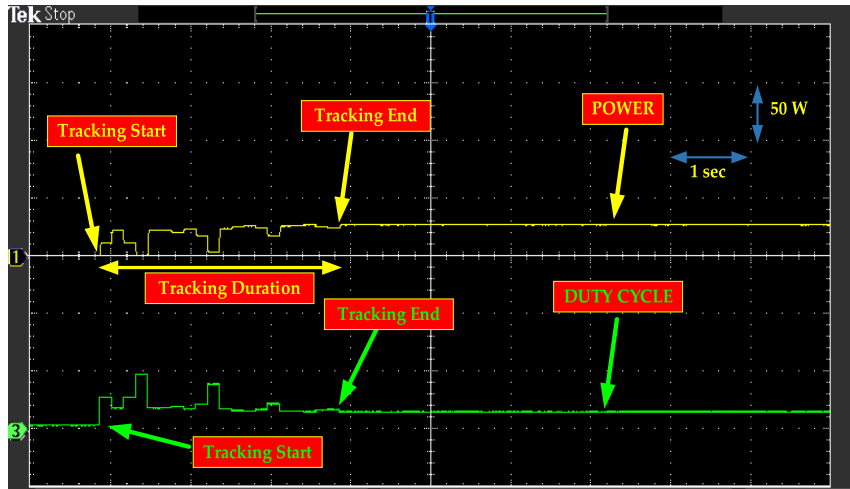


(b)



(c)

FIGURE 13. Comparison of the suggested algorithm's MPPT performance under condition 1 (1000, 600, 300, 100) W/m² with other algorithms: (a) PSA, (b) Cuckoo Search, (c) PSO, and (d) Jaya.



(d)

FIGURE 13. (Continued.) Comparison of the suggested algorithm's MPPT performance under condition 1 (1000, 600, 300, 100) W/m² with other algorithms: (a) PSA, (b) Cuckoo Search, (c) PSO, and (d) Jaya.

and

$$\left(\frac{\gamma(d_{ij}^{(I)})}{\gamma'(d_{ij}^{(I)})} \right) = \left(\frac{m \cdot d_{ij}^{(I)} - \frac{v^2}{2g} + \frac{g}{2v^2} \cdot (d_{ij}^{(I)})^2}{m + \frac{g}{v^2} \cdot d_{ij}^{(I)}} \right) \quad (19)$$

At the present iteration, the equation (20) formula is used to determine the distance between projectile and target:

$$d_{ij}^{(I)} = \left[\text{abs} \left((t_{ij}^{(I)})^r - (X_{ij}^{(I)})^r \right) \right]^{1/s} \quad (20)$$

The parameters r and s , where $(r, s) > 0$, are used to adjust the distance (d). The definition of another control parameter for (15) and (16) is shown below in equation (21).

$$\alpha = p^q, p = \left(1 - \frac{I}{I_{\max}} \right), q > 0 \quad (21)$$

Here, the new target population is additionally built as $t_{kj}^{(I+1)}$. The ideal projectile for the next iteration is calculated by using equations (22) and (23).

$$X_{bestj}^{(I+1)} = \begin{cases} X_{ij}^{(I+1)} & \text{if } \min(f(X_{ij}^{(I+1)})) \\ & \leq \min(f(t_{kj}^{(I+1)})) \\ t_{kj}^{(I+1)} & \text{otherwise} \end{cases} \quad (22)$$

$$f_{best}^{(I+1)} = f(X_{bestj}^{(I+1)}) \quad (23)$$

Until $f_{best}^{(I+1)}$ achieves the necessary accuracy level (ϵ) or the iteration reaches $I_{\max}$, the operations are repeated, with the end outcome being $(X_{bestj}^{(I+1)})$ as the pleased projectile touching down on the target.

Algorithm 1 displays the pseudocode for the suggested PSA algorithm.

The flowchart of the proposed algorithm has been shown in figure 9.

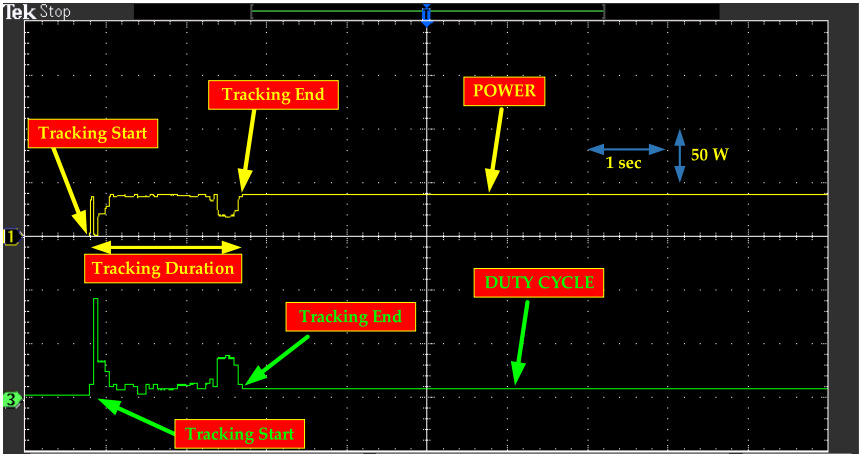
V. RESULTS AND DISCUSSIONS

The MPPT tracker architecture is shown in figure 10. It is a closed-loop controller with a driving circuit, sensors, a micro-controller with the MPPT algorithm, and a DC-DC boost converter. The microcontroller measures the voltage and current of the PV array to determine the Duty Ratio (D) value. The DC-DC boost converter generates the tuning signal using the values of D , which is then provided via the gate driver circuit. The controller's job is to carry out its MPPT algorithm, whose solutions are the best D values to extract P_{MP} from the PV array [32]. Equation (24) below defines the objective function (f) used in the MPPT problem:

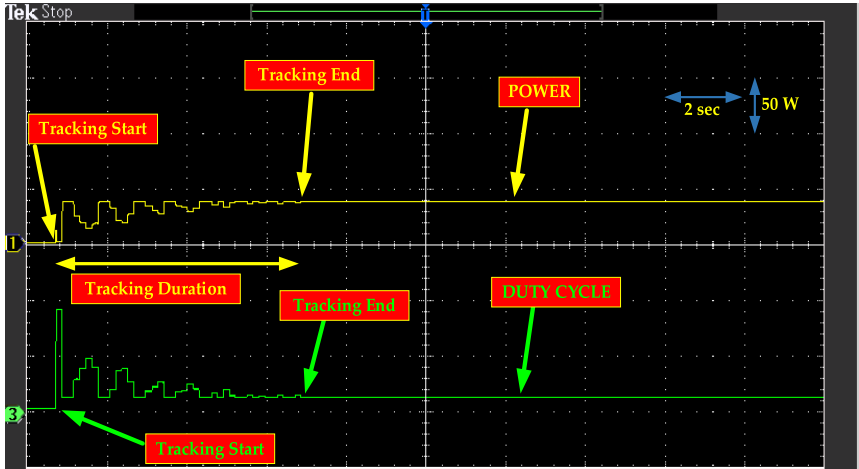
$$f(D) = P_{pv} \quad (24)$$

The study uses the Projectile Search Algorithm (PSA) to search for GMPP where a specific optimal value of D results in P_{MP} .

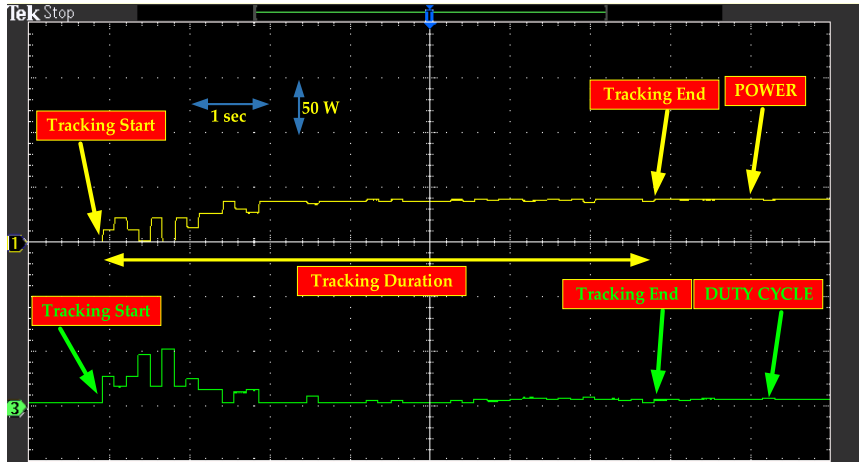
The following are the ratings of the different components that make up the boost converter: input capacitance (C_i) = 47 μF , inductance (L) = 1.1478 mH , output capacitance (C_o) = 470 μF , and output load (R) = 10 Ω . The solar photovoltaic array that is employed has a standard rating. The following is a list of the numerous parameters that may be found inside a module of an array: Maximum Power (P_{MPP}) is equal to 21.837 watts, Open Circuit Voltage (V_{OC}) is equal to 5.425 volts, Short Circuit Current (I_{SC}) is equal to 5.34 amperes, Voltage at Maximum Power (V_{MPP}) is equal to 4.35 volts, and Maximum Power Current (I_{MPP}) is equal to 5.02 amperes. Because the array that was selected had a total of four modules, its maximum power rating was calculated to be 87.348 watts. Following the completion of each cycle, the gate driver was provided with modulated pulses that included the current duty ratio. These pulses activated the switch,



(a)

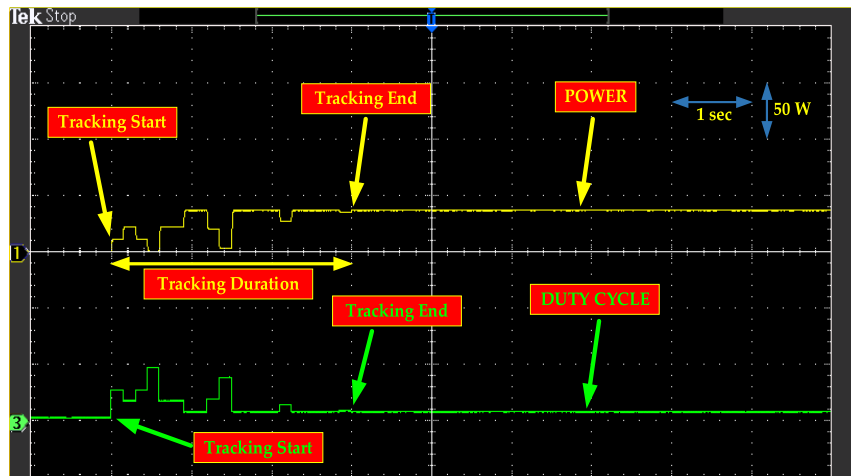


(b)



(c)

FIGURE 14. Comparison of the suggested algorithm’s MPPT performance under condition 2 (1000 900 500 400) W/m² with other algorithms: (a) PSA, (b) Cuckoo Search, (c) PSO, and (d) Jaya.



(d)

FIGURE 14. (Continued.) Comparison of the suggested algorithm's MPPT performance under condition 2 (1000 900 500 400) W/m² with other algorithms: (a) PSA, (b) Cuckoo Search, (c) PSO, and (d) Jaya.

TABLE 1. Control variables for the optimization method used in this work.

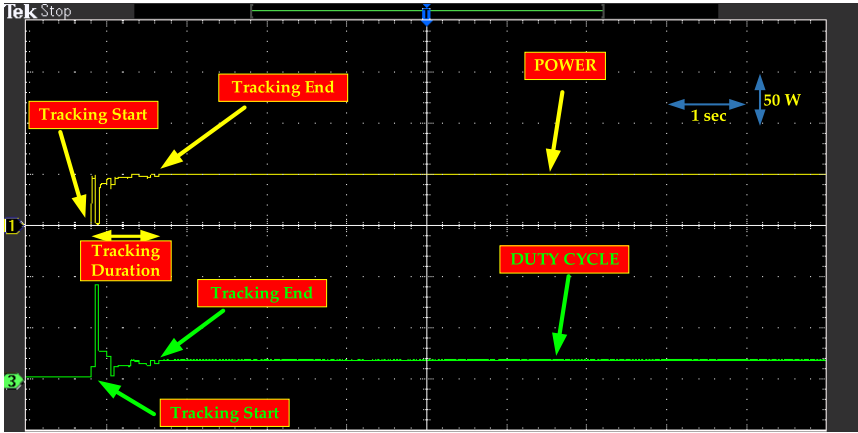
Optimization Techniques	Control Parameters	Values Used
PSA	Population Size (N), Maximum Iterations (T_{max}), Initial Velocity (V_i), Launch Angle (θ)	$N = 4$, $T_{max} = 100$, $V_i = 50$, $\theta = 45$
Cuckoo Search	Population Size (N), Parameter β , Discovery Probability (p_a), Maximum Iterations (T_{max})	$N = 4$, $\beta = 1.5$, $p_a = 0.3$, $T_{max} = 100$
JAYA	Population Size (N), Rate of Exploration (C_r), Rate of Learning (C_s), Maximum Iterations (T_{max})	$N = 4$, $C_r = 0.1$, $C_s = 0.9$, $T_{max} = 100$
PSO	Population Size (N), Maximum number of Iterations (T_{max}), Inertia Weight (w), Cognitive Factor (c_1), Social Factor (c_2)	$N = 4$, $T_{max} = 100$, $w = 0.7$, $c_1 = 1.5$, $c_2 = 2.0$

which then sent the output load the corresponding optimal power, the total of the optimized voltage and current. The recommended method will be examined alongside several additional state-of-the-art algorithms for a range of PSCs in the next section. The PSA for MPPT is evaluated using the Typhoon Hardware-In-Loop (HIL) emulator (402), as seen in Figure 11.

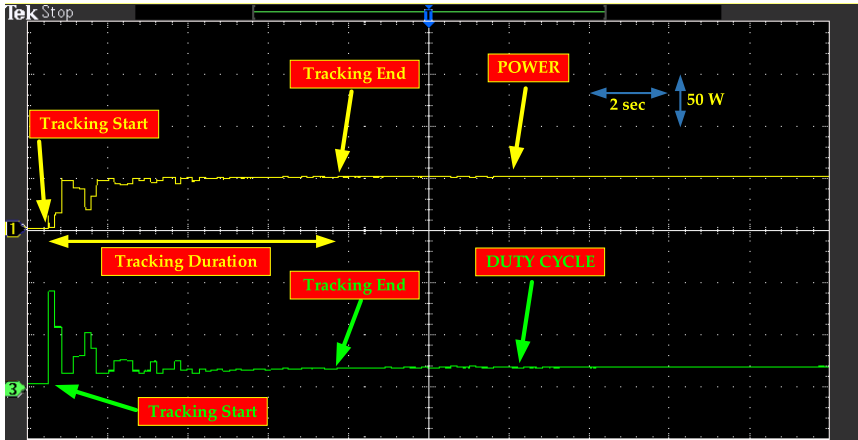
The control settings for the optimization methods used in this investigation are shown in Table 1. The methods that have been selected are Jaya, Cuckoo Search, PSA, and PSO. To direct the optimization process, certain control

parameters were applied to each approach. Table 1 also provides a summary of the stated values utilized for each parameter in this work, which makes it easier to replicate and understand the experimental settings for further evaluation and interpretation. The PV module used in this investigation has the specifications outlined in Table 2.

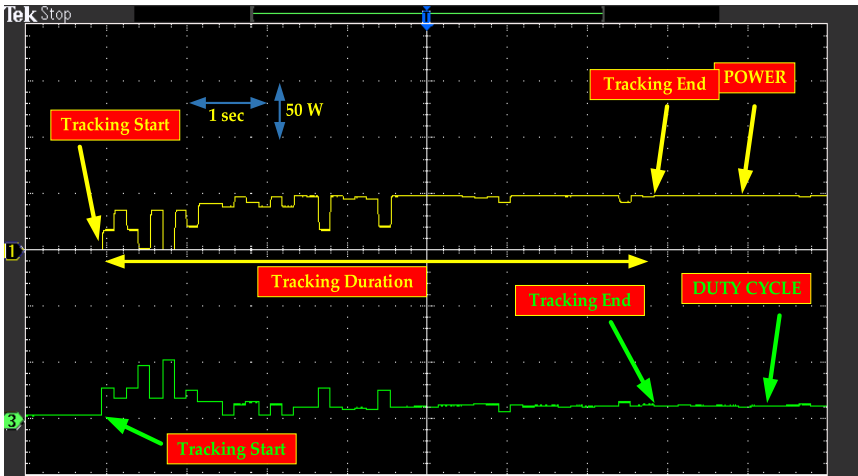
Table 3 displays distinct shading patterns on four solar PV modules together with the related insulation (W/m²) values for each panel under varied environmental circumstances. The various shade patterns used on the modules are referred to as conditions. For instance, in condition 1, panel 1 insulation



(a)

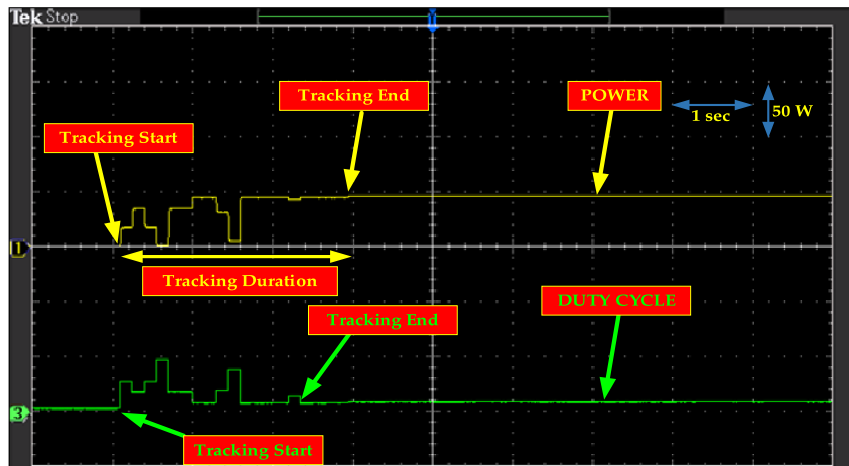


(b)



(c)

FIGURE 15. Comparison of the suggested algorithm's MPPT performance under condition 3 (1000 1000 1000 500) W/m² with other algorithms: (a) PSA, (b) Cuckoo Search, (c) PSO, and (d) Jaya.



(d)

FIGURE 15. (Continued.) Comparison of the suggested algorithm's MPPT performance under condition 3 (1000 1000 1000 500) W/m² with other algorithms: (a) PSA, (b) Cuckoo Search, (c) PSO, and (d) Jaya.

TABLE 2. Specifications of the PV module used in this investigation.

Configuration	Values
PV modules (series)	4 modules
PV cells per module	72 cells
Performance Metrics	
Open-circuit voltage (V_{oc})	5.425 Volts
Short-circuit current (I_{sc})	5.34 Amperes
Voltage at Maximum Power Point (V_{mpp})	4.35 Volts
Current at Maximum Power Point (I_{mpp})	5.02 Amperes
Maximum power output (P_{mpp})	21.837 Watts
Coefficients of Temperature	
Temperature coefficient of V_{oc} (K_v)	-0.37501 %/°C
Temperature coefficient of I_{sc} (K_i)	0.075 %/°C

is 1000 W/m², whereas it is 900, 500, and 400 W/m² on panels 2, 3, and 4, respectively. The insolation on panel 1 in condition is 1000, whereas it is 950, 700, and 650 W/m² on panels 2, 3, and 4, respectively. The remaining irradiance for conditions 3 and 4 at four different temperatures may be observed in Table 3. The way that the modules are shaded may have a significant impact on how well the solar PV system works. A solar PV module's power output may be decreased if any of its cells are shaded since less energy is produced overall.

Figure 12 illustrates the simulation assessments, including a thorough examination of various insolation parameters. As a result of this investigation, different forms and peaks were discovered in the photovoltaic (P-V) properties. Figures 13–16 display a comparison of the HIL results for the proposed PSA algorithm against other algorithms, including Cuckoo Search, PSO, and Jaya, under different shading

TABLE 3. Different patterns of shading on each of the four solar modules.

Condition	Insolation Values (W/m ²)			
	Panel 1	Panel 2	Panel 3	Panel 4
1	1000	600	300	100
2	1000	900	500	400
3	1000	1000	1000	500
4	1000	950	700	650

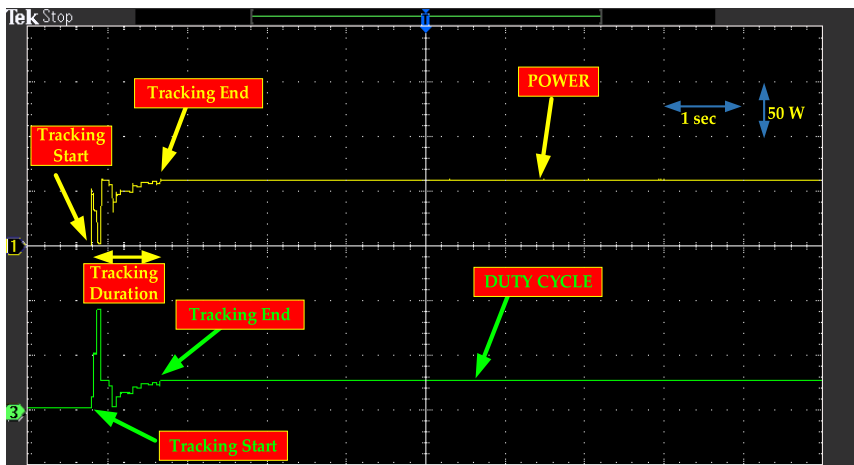
TABLE 4. Performance Comparison of the proposed MPPT algorithm (PSA) with the conventional MPPT algorithm (P&O).

Algorithms	PMPP	Efficiency	Tracking Time
PSA [P]	37.77 W	98.82 %	1.5 sec
P&O	21 W	54.94 %	0.5 sec

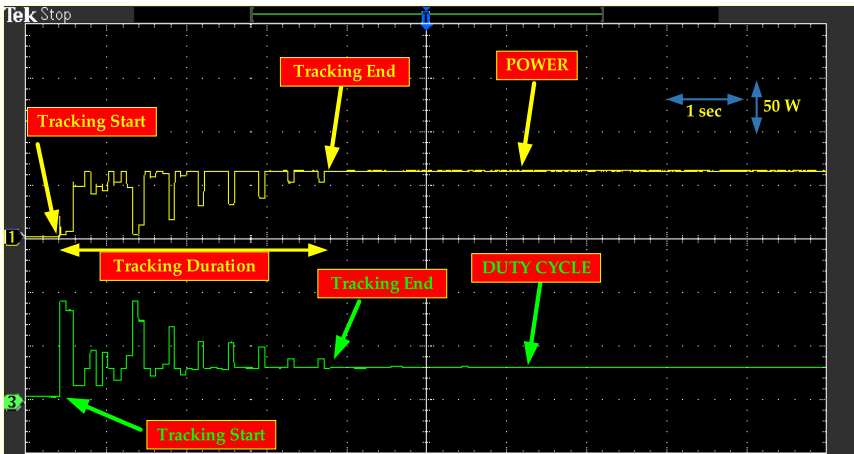
conditions. The Comparison charts illustrate the tracking times and efficiency of four different algorithms across four PSC conditions (Conditions 1, 2, 3, and 4) have been shown in figure 17. This graphic depiction makes it easy to compare PSA's tracking periods and operating efficiency to those of its competitors. Efficiency has been calculated using the formula:

$$\text{Efficiency} = \left(\frac{\text{Output power}}{\text{Input power}} \right) \times 100\% \quad (25)$$

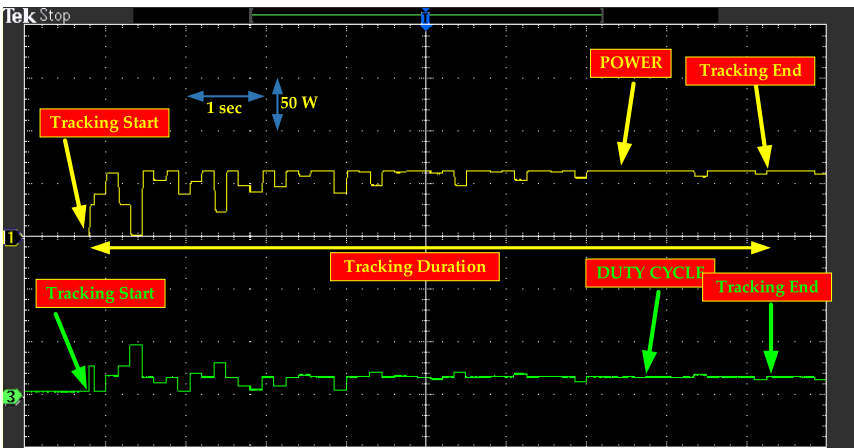
where output power is the tracked power and input power is the available PV power.



(a)

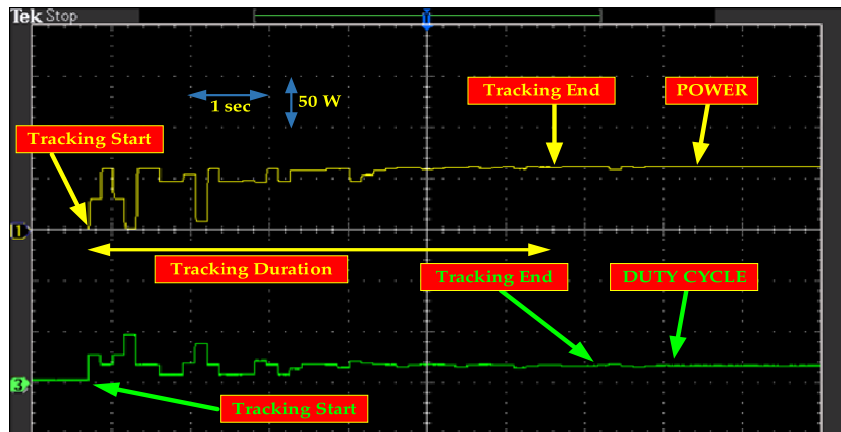


(b)



(c)

FIGURE 16. Comparison of the suggested algorithm's MPPT performance under condition 4 (1000 950 700 650) W/m² with other algorithms: (a) PSA, (b) Cuckoo Search, (c) PSO, and (d) Jaya.



(d)

FIGURE 16. (Continued.) Comparison of the suggested algorithm's MPPT performance under condition 4 (1000 950 700 650) W/m^2 with other algorithms: (a) PSA, (b) Cuckoo Search, (c) PSO, and (d) Jaya.

TABLE 5. Different insolation levels on the four modules of the PV array.

Scenario	Panel 1 (W/m^2)	Panel 2 (W/m^2)	Panel 3 (W/m^2)	Panel 4 (W/m^2)
Scenario 1	300	1000	1000	1000
Scenario 2	1000	400	1000	1000
Scenario 3	1000	1000	650	1000
Scenario 4	1000	1000	1000	500

Figure 18 compares the performance of the proposed MPPT algorithm with the conventional P&O algorithm under irradiance levels of (1000, 1000, 550, 200) W/m^2 .

Table 4 compares the performance of the proposed MPPT algorithm, Projectile Search Algorithm (PSA), with the conventional Perturb and Observe (P&O) algorithm across three key metrics: maximum power point (PMPP), efficiency, and tracking time. The results show that PSA algorithm achieves a significantly higher PMPP of 37.77 W, while P&O only reaches 21 W, indicating that PSA is more effective in optimizing power output from the solar panel. Additionally, PSA boasts an impressive efficiency of 98.82%, showcasing its capability to convert available solar energy into usable power, in stark contrast to P&O's much lower efficiency of 54.94%. Although P&O has a quicker tracking time of 0.5 seconds compared to PSA's 1.5 seconds, this rapid response can be attributed to P&O getting stuck in local maxima. Consequently, it may quickly settle on suboptimal power points, leading to its poor overall efficiency. In contrast, while tracking slower, PSA consistently finds the GMPP, justifying the longer tracking time with superior performance. These results suggest that PSA is the better choice for maximizing solar energy harvesting, especially in scenarios where efficiency is more critical than speed.

Figure 19 presents the HIL results for the proposed algorithm during dynamic changes in insolation, transitioning from full insolation of four PV panels (1000, 1000, 1000, 1000 W/m^2) to Scenarios 1, 2, 3, and 4 mentioned in Table 5. The figure illustrates the impact of dynamic insolation on the PV array, like how a cloud initially creates partial shading and then gradually moves away, revealing or obscuring the array's surface.

The results show that the suggested method performs the closest to the real performance while being fast, efficient, and reliable in static and dynamic condition both.

Table 6 illustrates the improvement in tracking time for the proposed PSA algorithm compared to three other metaheuristic algorithms, Jaya, PSO, and Cuckoo Search, across four different PSCs.

The table quantifies the percentage improvement in tracking time, highlighting the efficacy of the PSA algorithm. For instance, under Condition 1, PSA demonstrates a 50% improvement over Jaya, a 120% improvement over PSO, and a 240% improvement over Cuckoo Search. This trend continues through other conditions, with particularly significant improvements observed in Conditions 3 and 4. The data underscores the superior performance of the PSA algorithm in quickly reaching the maximum power point, thus enhancing the overall efficiency and responsiveness of the PV

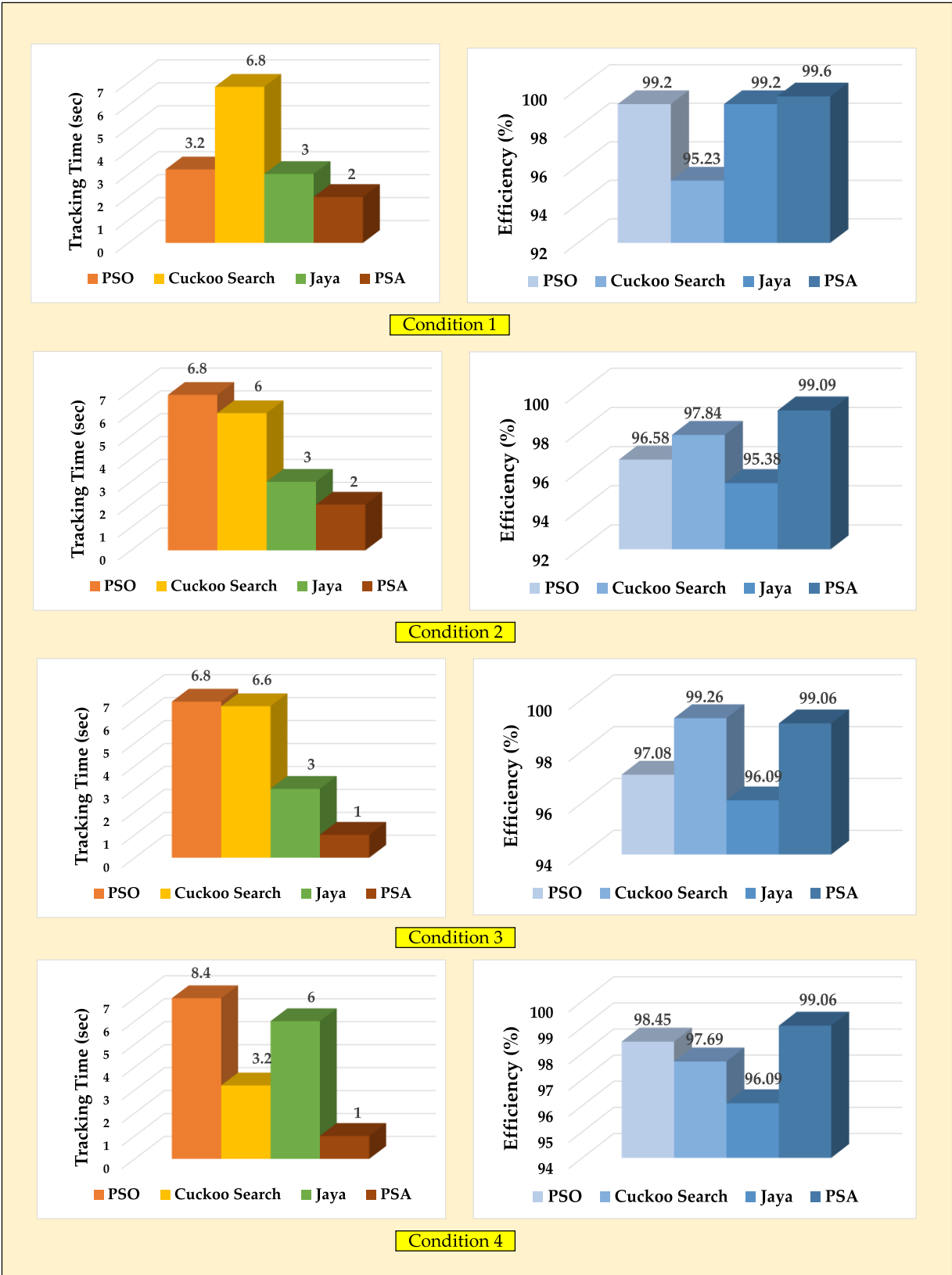
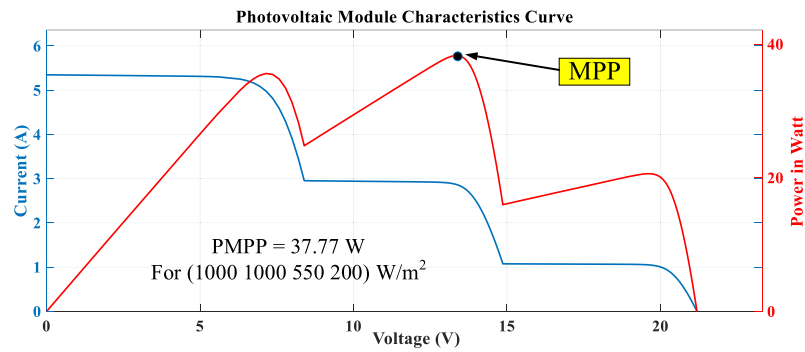
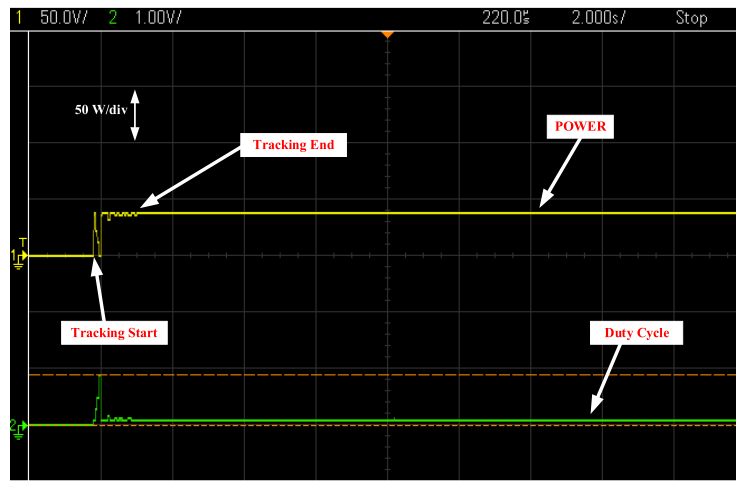


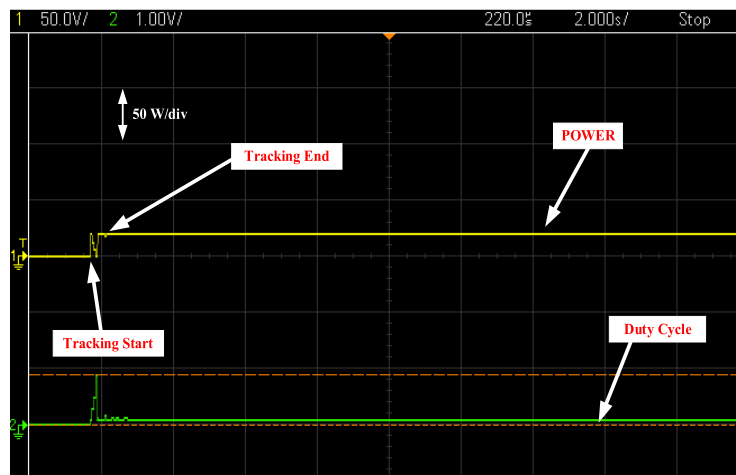
FIGURE 17. Comparison charts show the tracking times and efficiency of the four different algorithms for the four PSC (Conditions 1, 2, 3, and 4).



(a)



(b)

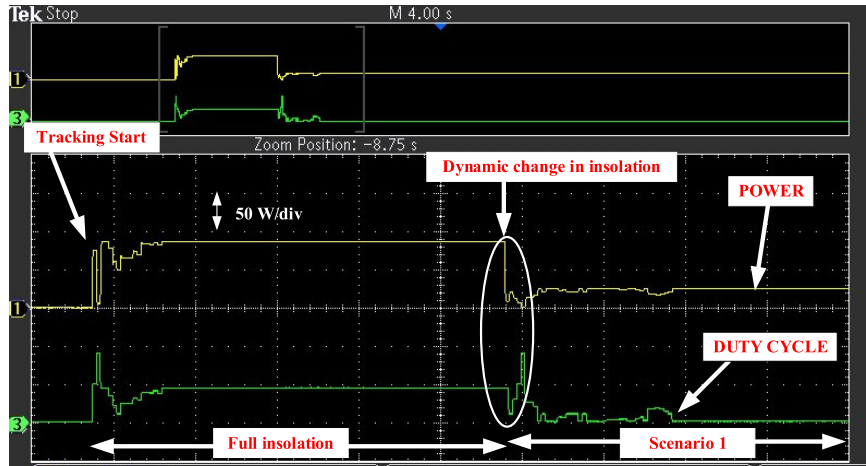


(c)

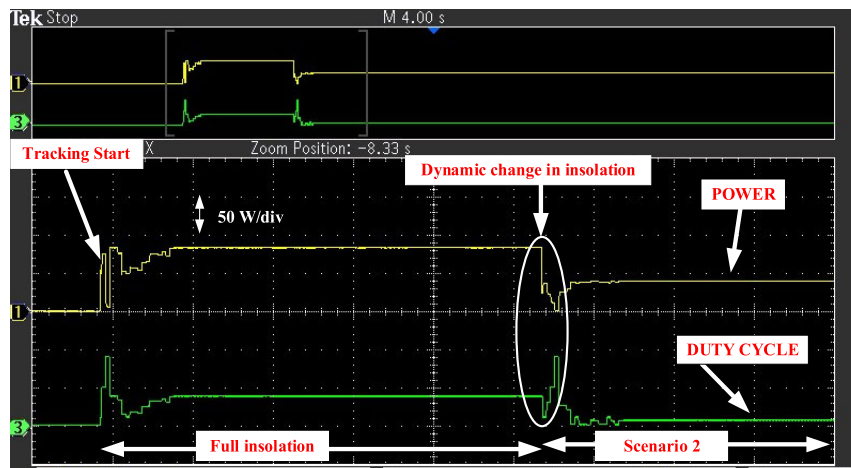
FIGURE 18. Comparison of the proposed algorithm's MPPT performance with the conventional P&O algorithm under irradiance levels of (1000, 1000, 550, 200) W/m²: (a) P-V characteristics curve (b) PSA, (b) P&O.

system under varying shading conditions. This comparison validates the effectiveness of the PSA algorithm in optimizing

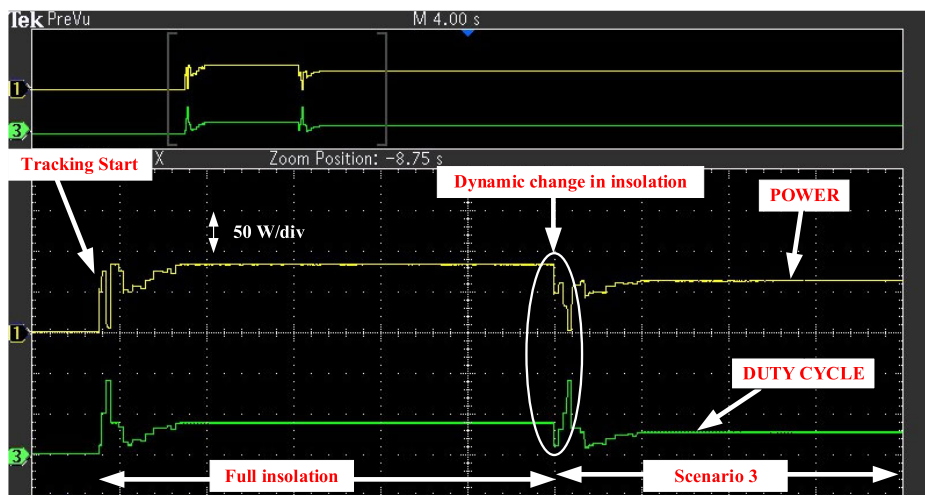
the performance of PV systems, making it a compelling choice for real-world applications. The real-time HIL data for



(a)



(b)



(c)

FIGURE 19. HIL results of the PSA-based MPPT during dynamic insolation changes: (a) from full insolation ($1000, 1000, 1000, 1000 \text{ W/m}^2$) to ($300, 1000, 1000, 1000 \text{ W/m}^2$), (b) from full insolation to ($1000, 400, 1000, 1000 \text{ W/m}^2$), (c) from full insolation to ($1000, 1000, 650, 1000 \text{ W/m}^2$), and (d) from full insolation to ($1000, 1000, 1000, 500 \text{ W/m}^2$).

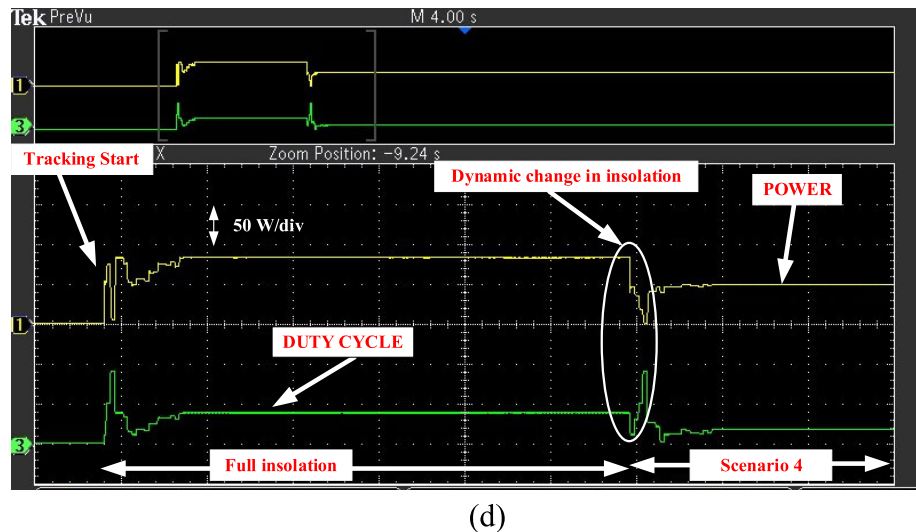


FIGURE 19. (Continued.) HIL results of the PSA-based MPPT during dynamic insolation changes: (a) from full insolation (1000, 1000, 1000, 1000 W/m²) to (300, 1000, 1000, 1000 W/m²), (b) from full insolation to (1000, 400, 1000, 1000 W/m²), (c) from full insolation to (1000, 1000, 650, 1000 W/m²), and (d) from full insolation to (1000, 1000, 1000, 500 W/m²).

TABLE 6. Calculate tracking time improvement as a percentage and compare the four techniques.

Different PSC	Improvement in Tracking Time (%) - PSA vs. Jaya, PSO and Cuckoo search		
	Jaya	PSO	Cuckoo search
Condition 1	50 %	120 %	240 %
Condition 2	50 %	240 %	200 %
Condition 3	200 %	580 %	560 %
Condition 4	500 %	740 %	220 %

different shading conditions reveals an important conclusion. All these results support the idea that the proposed PSA algorithm outperforms the PSO, Jaya, and Cuckoo search algorithms in several important performance areas.

In summary, when comparing the convergence times across the four states, it is evident that PSA consistently outperforms the other algorithms. It consistently achieves faster convergence times, reaching the optimal solution in the shortest duration. Cuckoo Search, PSO, and JAYA demonstrate relatively longer convergence times across the states, indicating that more time is required to converge towards the optimum solution. Therefore, according to the provided results, PSA shows promise as an efficient algorithm for optimization problems, providing quick convergence and potentially saving computational time and resources.

As shown in Figure 12(a–d), the suggested method demonstrated its capacity to hold solutions within a limited search

area by settling on the MPP with a quick rate of convergence and no significant size oscillations. As a result, the algorithm was able to show that it could maintain solutions within a limited search area. Since the initialization of the duty ratio causes the first substantial fluctuation, which does not influence how well the algorithm works, this variation is not included as a fluctuation since it is irrelevant to the algorithm's performance. Another benefit of limited, intelligent diversity is that, if the MPP is adhered to, the eventual convergence option may be reached more rapidly, thereby saving additional power losses. This is possible if the diversification is kept relatively small. All these elements assist in a considerable increase in the PV total efficiency array by helping to utilize the electricity produced by the array.

VI. CONCLUSION

This study aims to provide an effective Projectile Search Algorithm (PSA)-based MPPT controller for a solar PV system. Several metaheuristic algorithms, including PSO, Jaya, and Cuckoo search, are compared with the proposed PSA-based MPPT method under various shading circumstances. Utilizing the Typhoon hardware-in-the-loop (HIL)-emulator (402), analyses the PV system in real time and assesses the effectiveness of the suggested method in static and dynamic conditions. According to the findings, the MPPT controller that uses the PSA performs better than other algorithms regarding tracking speed, power variations, and efficiency. The findings further demonstrate that the MPPT algorithm based on PSA can track at a high rate of convergence and consistently achieve the GMPP. It also stays clear of large oscillations, which may reduce efficiency and result in power losses. With an average efficiency of 99.06%, the PSA technique takes an average of

1.5 seconds to converge on the GMPP. For these reasons, PSA could be an optimum algorithm for creating durable, reliable, and efficient MPPT controllers for real-world use.

ACKNOWLEDGMENT

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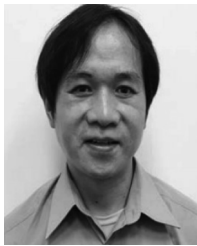
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